

From Explanation to Action: A Case Study of Agronomist Workflows

Rui Pedro Porfírio
NOVA LINES
NOVA School of Science and
Technology, NOVA University of
Lisbon
Caparica, Portugal
Setúbal School of Technology,
Polytechnic University of Setúbal
Setúbal, Portugal
r.porfirio@campus.fct.unl.pt

Pedro Albuquerque Santos
NOVA LINES
NOVA School of Science and
Technology, NOVA University of
Lisbon
Caparica, Portugal
Setúbal School of Technology,
Polytechnic University of Setúbal
Setúbal, Portugal
pedro.albuquerque@estsetubal.ips.pt

Rui Neves Madeira
NOVA LINES
NOVA School of Science and
Technology, NOVA University of
Lisbon
Caparica, Portugal
Setúbal School of Technology,
Polytechnic University of Setúbal
Setúbal, Portugal
rui.madeira@estsetubal.ips.pt

Abstract

Agronomists bear the responsibility of verifying AI predictions to ensure safe agricultural decision-making. However, a critical gap persists in understanding how these domain experts leverage Explainable AI (XAI) to verify diagnoses within their workflows. We conducted a qualitative case study with nine professional agronomists to evaluate feature-attribution and example-based explanations. Our observations suggest that abstract heatmaps risk reinforcing automation bias by obscuring the biological ground truth. In contrast, example-based explanations appeared to support the experts' epistemic practice of situated seeing. Furthermore, participants consistently prioritized actionability over transparency. They viewed the diagnosis not as an endpoint, but as a prerequisite for intervention. Consequently, this work contributes design considerations to bridge the gap between static model explanations and active agronomic scrutiny.

CCS Concepts

• **Applied computing** → **Agriculture**; • **Computing methodologies** → *Artificial intelligence*; • **Human-centered computing** → **Empirical studies in HCI**; *Interaction design*.

Keywords

Explainable AI, Human-centered AI, Qualitative Study, Agronomy, Domain Experts, Digital Agriculture

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1 Introduction

The challenge of ensuring global food security while meeting urgent sustainability goals has placed immense pressure on agriculture.

This has resulted in a shift towards digital agriculture, where data-driven technologies are used to enhance farming efficiency [35]. At the core of this movement are predictive systems intended to support tasks such as disease diagnosis. However, the success of these systems depends on whether they support expert verification within socio-technical workflows [9]. Expert agronomists are tasked with interpreting these outputs to guide on-farm decisions. However, opaque "black box" systems create barriers to their work and contribute to low adoption rates [6]. This challenge is critical because agronomists bridge the gap between technology and practice. The emergence of concepts such as Human-Centered XAI (HCXAI) aims to make machine learning systems understandable and practically useful for such experts [2, 10, 11]. While this lens has proven crucial in healthcare and legal domains [19, 25, 31], XAI-related research in agriculture remains predominantly algorithm-centric [16, 29, 32].

To address this gap, we must examine how experts interact with common post-hoc explanation methods. Complex models like deep neural networks often rely on explanations generated after training [23]. Feature-attribution explanations, such as Grad-CAM saliency maps, highlight critical input regions the model deemed most important for its prediction [30]. These methods have the potential to help experts verify model reasoning against their own domain knowledge [22]. In contrast, example-based explanations select representative data instances to support analogical reasoning [2, 12]. Prior Human-AI Interaction research reveals established trade-offs between these strategies. Feature attribution methods can guide attention but may hinder human-AI team performance in difficult tasks [1, 24]. Conversely, example-based explanations are often more intuitive for building comprehension, but they risk causing confusion if the examples appear logically inconsistent [4, 18].

While system transparency supports trust and technology adoption [34], low confidence in digital reliability continues to limit smart farming [17]. This challenge is particularly acute in the Global South, where technology adoption rates remain limited [5]. However, for domain experts, existing knowledge enables them to filter and dismiss flawed recommendations [21, 28]. This role positions the predictive system as a "second opinion" requiring rigorous verification [7]. Consequently, research has shown that experts should employ scrutiny to guard against automation bias through contestability [33]. This engagement aligns with the concept of "situated seeing" [20], a process in which experts layer digital representations



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onto their embodied local knowledge to form a diagnosis. Hence, we conducted a case study with nine professional agronomists using a misleading diagnosis to evaluate how different explanations support error detection.

This case study is organized into four primary sections. Section 2 describes the methodology, including recruitment and the design of the misleading diagnosis scenario. Section 3 presents qualitative observations that contrast abstract visualizations with analogical verification. Furthermore, Section 4 discusses design considerations for contestable diagnostic tools. Finally, Section 5 explores broader implications for human-centered AI and concludes the work.

2 Methodology

We employed a qualitative approach to capture the nuance of expert reasoning within diagnostic workflows. This methodology integrated a think-aloud protocol with a subsequent co-design task to study participant interactions. Furthermore, we used quantitative measures to triangulate the primary qualitative observations. The following subsections detail the participant cohort, the experimental prototype, and the study procedure.

2.1 Participants and Design

We employed a within-subjects design with nine Portuguese professional agronomists to gather comparative feedback while controlling for individual differences in domain knowledge. Given our qualitative focus on depth over breadth, this sample size is consistent with established findings showing that thematic saturation is often reached within 9 to 17 interviews for studies with homogenous populations [13]. The group had extensive professional experience in agronomy with an average of 7.3 years in relevant roles. Furthermore, the educational background of the cohort was uniformly high. All nine participants held a university degree. Participants reported high familiarity with general AI-based tools but moderate-to-low familiarity with XAI concepts. This strong domain expertise combined with limited prior exposure to XAI, supported their suitability for researching the initial perceptions of these novel systems. The presentation of the three conditions was counterbalanced, using a balanced Latin Square design, to ensure rigor and mitigate ordering effects.

2.2 Prototype and Conditions

The study used an interactive web application prototype localized in Portuguese. This system simulated a deep learning tool for plant disease diagnosis. It was developed as a functional proof-of-concept building on the AgriUXE platform [27]. All explanations were pre-generated and cached for the sessions to ensure a consistent user experience. For each task, participants assessed one of three experimental conditions as shown in Figure 1. The Baseline (C1) presented only the model's diagnosis. The Feature-Attribution condition (C2) added a Grad-CAM saliency map to show which visual features the model deemed most important. Conversely, the Example-Based condition (C3) displayed two visually similar and confirmed images from the training dataset. These instances were retrieved via a k-Nearest Neighbors (k-NN) search on feature embeddings extracted from the system's pre-trained Vision Transformer model [8]. This

approach ensured that the examples shared high visual and semantic similarity with the input image to support analogical reasoning.

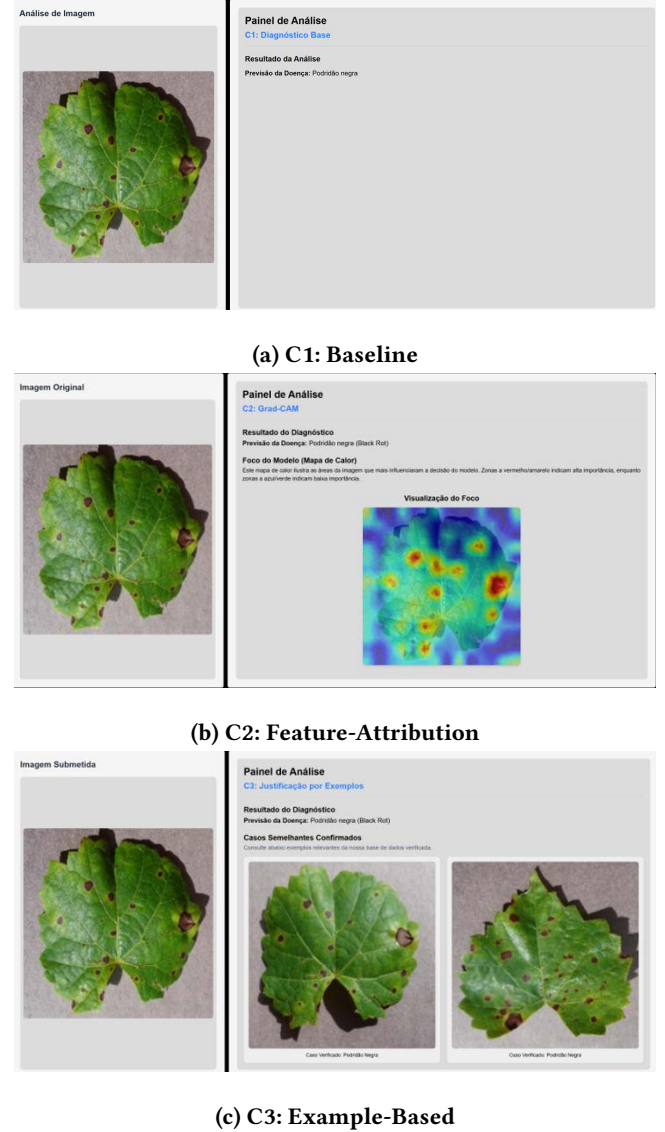


Figure 1: The three experimental conditions presented to participants. (a) The Baseline provided the diagnosis and confidence. (b) The Feature-Attribution added a Grad-CAM heatmap. (c) The Example-Based added two similar, confirmed cases.

2.3 Study Procedure

We used three plant images from different crops as experimental inputs. These included a grape leaf with black rot and a potato leaf with leaf blight. However, the last image was a healthy apple leaf which was central to our verification scenario. For this image, the machine learning model incorrectly classified the leaf with a

diagnosis of apple scab. We used this misleading diagnosis to specifically assess how each explanation condition supported an expert's ability to detect a flawed output. Each session lasted approximately 45 minutes and was audio and video recorded. Participants engaged with the prototype while adhering to a think-aloud protocol. Facilitator intervention was kept to a minimum to encourage continued verbalization. Upon completing each task, they filled out a post-task questionnaire to record quantitative measures such as trust and cognitive load. The final phase featured a brief semi-structured interview followed by a generative co-design task. This task centered on a purpose-built UI construction tool as seen in Figure 2. This tool provided a palette of pre-defined components that participants could drag onto a canvas. Furthermore, a separate drawing tool was available to create novel interface elements.

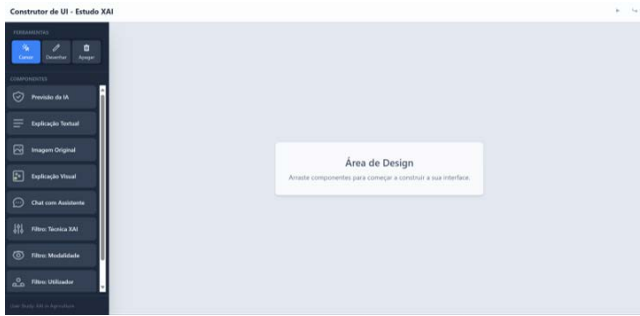


Figure 2: The interface for the co-design task.

A diverse set of data sources were collected to allow for methodological triangulation. Quantitative measures included post-task scores from the Trust in Explainable AI (TXAI) scale [15], the PaaS Mental Effort Rating Scale [26] for cognitive load, and the Explanation Satisfaction Scale (ESS) [15]. The analysis followed a sequential explanatory strategy. Descriptive statistics were generated from the questionnaires to contextualize the qualitative findings. This approach uses quantitative data to add precision to the story rather than for making claims of statistical significance. Finally, the thematic analysis followed the reflexive approach of Braun and Clarke [3] with regular meetings to cluster codes into synthesized themes.

3 Observations

The quantitative data revealed a clear consistent preference for example-based explanations (C3). As summarized in Table 1, this condition gathered the highest scores for decision confidence and trust. Furthermore, it imposed the lowest reported cognitive load. However, these descriptive metrics do not fully capture the complexities of expert verification. Data from the misleading diagnosis scenario revealed an interesting gap in error detection across conditions. Notably, no participant using the heatmap condition rejected the wrong prediction. In contrast, one participant in the example-based group successfully identified the error. This preliminary finding highlights a potential divergence in how experts interact with different explanation modalities. However, given the small sample size ($N = 3$ per condition) in this specific task, these

patterns should be treated as anecdotal observations rather than statistical evidence.

Metric	C1: Baseline	C2: Heatmap	C3: Examples
Mean Decision Confidence	5.1	5.6	5.9
Mean Compounded Trust	3.65	3.89	3.98
Mean Cognitive Load	2.78	2.89	2.56
1st Place Preference	0	2	7

Table 1: Descriptive metrics ($N=9$). Examples (C3) scored highest in confidence and trust.

3.1 Visual Abstraction and Verification

The Feature-Attribution condition (C2) presented a contradiction between high reported trust and low error detection. P8 misinterpreted the red regions of the Grad-CAM heatmap as literal temperature readings. This suggests that the visualization conveyed an unearned sense of scientific precision. Consequently, this perceived authority may have reinforced automation bias. In the verification scenario, none of the participants in this condition challenged the misleading diagnosis. This implies that the abstract visualization may have discouraged experts from questioning the machine's output.

3.2 Analogical Reasoning and Scrutiny

In contrast, the Example-Based condition (C3) aligned with the experts' epistemic practice of analogical reasoning. Participants described this approach as a *"natural way"* (P4) to obtain a *"second opinion"* (P6). This alignment appeared to support active verification in specific instances. For example, P1 successfully rejected the misleading diagnosis by using the explanation as evidence. The participant noted: *"I have little confidence... because it seems to me that the initial image is the same as one of the [healthy] examples"*. While this represents a single observation, it offers preliminary indication that grounding AI outputs in visual comparisons may help experts apply their professional vision. However, the participants who failed to detect the error in this condition appeared to align the visual evidence with the model's label. For instance, when analyzing the misleading diagnosis, P4 justified the AI decision by stating: *"I think the other one (the healthy image) is a little bit dry. That might be the part that is the pathology."*

3.3 Actionability and Expert Workflow

Finally, the analysis revealed that our participants did not view a diagnosis as an endpoint. Instead, they treated it as a prerequisite for intervention. P8 stated explicitly: *"The next step, after knowing the disease, is to know how to cure it"*. P1 echoed this sentiment: *"The disease is identified, now what are we going to do?"*. Therefore, perceived value in this domain seems to be driven by actionability. Experts consistently demanded that the system support the workflow from detection to treatment, rather than providing a static label in isolation.

4 Design Considerations

Our findings advocate for a shift from model-centric transparency to situated expert agency. We propose four considerations to support the professional workflows of agronomists.

4.1 Design for Scrutiny and Contestability

Experts do not treat a predictive system as a certain oracle. Instead, they view it as a colleague subject to verification. Aligning with Hirsch et al.'s concept of contestability [14], our preliminary findings show a consistent need for systems designed as interactive analytical environments that empower users to be active interrogators. This requires providing tools to assess the primary evidence first, allowing users to form their own opinion before being influenced by the prediction. Furthermore, during the co-design task, the participants actively highlighted the value of conversational tools (e.g., a chatbot) as enabling users to ask contrastive questions and request alternative evidence when a diagnosis is suspect.

4.2 Align with Temporal and Environmental Context

Agronomic reasoning requires ecological validity. Therefore, a contestable system must allow experts to verify predictions against seasonal cycles and local weather patterns. This aligns with the expert demand for holistic evidence. As P3 noted, relying on a single data point is insufficient: *"Normally with just one leaf, it is difficult to reach a diagnosis. It would be important to have an image of how the tree looks"*. Consequently, systems should enable the integration of broader environmental context to reflect the situated reality of the field.

4.3 Support Situated Seeing through Analogical Reasoning

Participants often preferred analogical reasoning over abstract visualizations of model internals. Saliency maps frequently led to ambiguity or misinterpretation. In contrast, verified examples are more intuitive for verification tasks. This support for situated seeing is most effective when systems provide multimodal justifications [20]. For instance, P2 suggested that textual justifications should be confirmed by visual proof.

4.4 Support Accountability through Actionability

The expert journey begins with the diagnosis rather than ending there. Consequently, systems must bridge the gap from prediction to intervention. This requires integrating actionable recommendations for treatment or prevention. Without this pathway, the system fails to support the professional accountability of the agronomist. Actionable guidance ensures the tool remains a starting point for practical on-farm decisions.

5 Discussion and Conclusion

The findings of this study highlight a misalignment between the current paradigm of XAI and the situated reality of agricultural work. While model-centric explanations, such as heatmaps, can

be technically transparent, our results suggest they may fail to withstand the practical scrutiny required in high-stakes domains.

The divergence observed in our verification scenario serves as a critical observation for the human-centered computing community. The feature-attribution condition (C2) garnered high trust scores. However, it failed to enable participants in this group to detect the misleading diagnosis. This suggests that abstract visualizations may create an illusion of understanding. Such visuals convey an unearned scientific authority that effectively discourages questioning. In contrast, the example-based condition (C3) appeared to break this pattern of passive acceptance. By grounding the machine learning output in comparative evidence, this condition may enable users to leverage their domain knowledge to successfully challenge the system. Although these patterns remain anecdotal due to the small sample size, they suggest that trust is a potentially dangerous metric for safety in expert domains. Design efforts should prioritize contestability. This principle describes the degree to which an interface empowers the user to identify and reject false positives [14].

Furthermore, our analysis suggests that existing contestability frameworks must be extended to account for the temporal nature of agronomy. Existing frameworks often focus on episodic tasks where the relevant evidence is contained within a single interaction. However, agricultural reasoning requires establishing ecological validity over time. Agricultural experts do not make isolated choices. They contest predictions against seasonal disease cycles and regional environmental variations. Therefore, an effective tool must support reasoning that extends beyond the pixel data of a single image to include the broader disease lifecycle. These preliminary findings suggest that low adoption rates of smart farming technologies may stem from a fundamental mismatch between static AI outputs and situated expert workflows. While our results are exploratory, they indicate that future design must prioritize interactive analytical environments. Treating the AI diagnosis as an invitation for inquiry ensures that technology supports the situated seeing where experts layer digital data over their professional vision [20].

Finally, we argue that the goal of XAI in agriculture must evolve toward enabling a broader process of interactive, evidence-based inquiry. By focusing on the specific epistemic needs of agronomists, this work contributes considerations for designing XAI systems that move beyond transparency to become contestable and verifiable in the real world.

We acknowledge that these insights are derived from a small sample of nine Portuguese experts. Consequently, these findings are exploratory and should be interpreted as hypotheses for future investigation rather than definitive proofs. It is also important to note that the verification scenario used a single case where participants sometimes lacked crop-specific domain knowledge. This limits the generalizability of our error detection findings regarding other crops. Future work should build upon these findings to evaluate fully functional prototypes with a larger and more diverse international sample. This will enable researchers to determine if thematic saturation was reached and to explore the potential transferability of these considerations to other high-stakes domains.

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